SOMAIYA VIDYAVIHAR UNIVERSITY

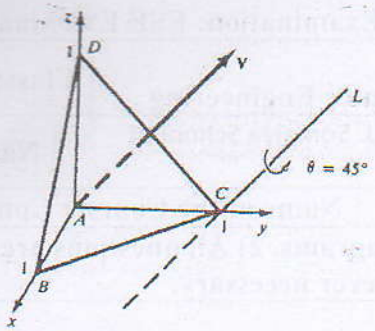
24.05.2025 (E)

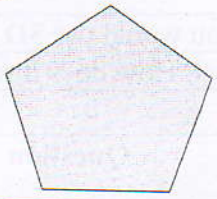
Semester: January 2025 –April 2025		
Maximum Marks: 100 **Examination: ESE Examination**		**Duration:3 Hrs.**
Programme code: 01 **Programme: BTech in Computer Engineering**	**Class:** TY	**Semester:(SVU 2020)** V
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** COMP	
Course Code: 116U01E511	**Name of the Course: Computer Graphics**	
Instructions: 1) Draw neat diagrams. 2) All questions are compulsory. **3) Assume suitable data wherever necessary.**		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Explain 2D pipeline in computer graphics with diagram	5
ii)	In the context of computer game development, how might the choice of visible surface detection method impact gameplay and user experience?	5
iii)	What is a clipping window in computer graphics, and how does it relate to the clipping process?	5
iv)	Describe what viewport transformations are and why they are used in computer graphics pipelines.	5
v)	Provide an example of when you would use 3D scaling in 3D modeling.	5
vi)	Explain the additive color model. How does it relate to the RGB color space?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	How are Euler angles used to represent 3D rotations? What are their advantages and limitations?	5
ii)	Describe different types of light sources used in computer graphics. How does each type affect the rendering of 3D scenes?	5
	OR	
Q2 A	List and briefly describe the primary geometric transformation functions available in OpenGL for 3D transformations.	10
Q 2 B	Solve any **One**	**10**
i)	What is event-driven programming, and how does it relate to capturing and responding to user input in interactive graphics applications? Provide the OpenGL functions used for event driven programming.	10
ii)	What steps are required to plot a line whose slope is between 0° and 45° using Bresenham's method and DDA method?	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Magnify the triangle with vertices A(0, 0), B(1, 1), and C(5,2) to thrice its size while keeping B fixed.	10
ii)	Show that the order in which transformations are performed is important by the transformation of triangle A(1, 0), B(0, 1), C(l, 1), by (a) rotating 45° about the origin and then translating in the direction of vector I, and (b) translating and then rotating.	10
iii)	The pyramid denned by the coordinates, A(0, 0, 0), B(1, 0, 0), C(0, 1, 0), and D(0,	10

	0, 1) is rotated 45° about the line L that has the direction V = J + K and passing through point C(0, 1, 0) (Fig). Find the coordinates of the rotated figure. 	

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	(a) Find the matrix that represents rotation of an object by 35° about the origin. (b) What are the new coordinates of the point P(4, -5) after the rotation?	10
ii)	Write a Pseduocode for drawing the Bezier curve. Provide the OpenGL functions used for same.	10
iii)	Write a Pseduocode for drawing following shape given in figure. Provide the OpenGL functions used for same. 	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	Write a short note on homogeneous coordinates.	5
ii)	Write a short note on 2DComposite transformations.	5
iii)	Explain aliasing and anti-aliasing effect in computer graphics.	5
iv)	What is 2D point clipping, and why is it necessary in computer graphics?	5
v)	How can you specify the position and characteristics of light sources in OpenGL?	5
vi)	Explain the concept Z-buffer; and its role in implementing the depth buffer method for visible surface determination?	5